2201NSC Linear Algebra and Applications

Week 6

David Harman

Inner Product Spaces

Length, orthogonality, similarity and best approximation

An inner product compares two vectors

An **inner product** takes two vectors $\mathbf{x}$ and $\mathbf{y}$ and returns a real number:

$$\boxed{\langle \mathbf{x}, \mathbf{y} \rangle \in \mathbb{R}.}$$

It provides a way to extend familiar geometric ideas to more general vector spaces.

Length

$$\|\mathbf{x}\|^2 = \langle \mathbf{x}, \mathbf{x} \rangle$$

Orthogonality

$$\mathbf{x} \perp \mathbf{y} \iff \langle \mathbf{x}, \mathbf{y} \rangle = 0$$

The dot product is an example of an inner product

For $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, the **standard inner product** is the dot product:

$$\boxed{\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x} \cdot \mathbf{y} = \mathbf{x}^{\mathsf{T}} \mathbf{y} = x_1 y_1 + \cdots + x_n y_n.}$$

It also satisfies

$$\mathbf{x} \cdot \mathbf{y} = \|\mathbf{x}\| \, \|\mathbf{y}\| \cos \theta.$$

If $\mathbf{x} \cdot \mathbf{y} = 0$, then $\mathbf{x}$ and $\mathbf{y}$ are **orthogonal**.

An inner product defines the norm of a vector

The norm, or length, of $\mathbf{x}$ is defined by

$$\boxed{\|\mathbf{x}\| = \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle}.}$$

Using the standard inner product,

$$\|\mathbf{x}\|^2 = \langle \mathbf{x}, \mathbf{x} \rangle = \mathbf{x}^\mathsf{T}\mathbf{x} = x_1^2 + \cdots + x_n^2.$$

For example, if $\mathbf{x} = (3, 4)^\mathsf{T}$, then

$$\|\mathbf{x}\| = \sqrt{3^2 + 4^2} = 5.$$

An inner product satisfies three properties

An operation $\langle \mathbf{x}, \mathbf{y} \rangle$ is an inner product when, for all vectors and real scalars, it satisfies:

1. **Positive definiteness**

$$\langle \mathbf{x}, \mathbf{x} \rangle \geq 0, \quad \text{with equality iff } \mathbf{x} = \mathbf{0}.$$

2. **Symmetry**

$$\langle \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{y}, \mathbf{x} \rangle.$$

3. **Linearity**

$$\langle \alpha \mathbf{x} + \beta \mathbf{y}, \mathbf{z} \rangle = \alpha \langle \mathbf{x}, \mathbf{z} \rangle + \beta \langle \mathbf{y}, \mathbf{z} \rangle.$$

Weighted inner product

For positive weights w_i, the rule

$$\langle \mathbf{x}, \mathbf{y} \rangle_w = \sum_{i=1}^{n} w_i x_i y_i$$

defines a weighted inner product.

For example, on $\mathbb{R}^2$ let

$$\langle \mathbf{x}, \mathbf{y} \rangle_w = 2x_1 y_1 + x_2 y_2.$$

Inner products also compare functions

For continuous functions on $[a, b]$, define

$$\boxed{\langle f, g\rangle = \int_a^b f(x)g(x)\,\mathrm{d}x.}$$

On $[0, 1]$,

$$\langle x, \sin(\pi x)\rangle = \int_0^1 x\sin(\pi x)\,\mathrm{d}x = \frac{1}{\pi}.$$

The same geometry applies to functions

Vectors need not be arrows or lists of numbers. Inner products provide the same ideas of length, orthogonality and similarity in spaces of functions.

Orthogonal and orthonormal sets

A set $\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ is **orthogonal** when

$$\langle \mathbf{v}_i, \mathbf{v}_j \rangle = 0 \qquad (i \neq j).$$

It is **orthonormal** when it is orthogonal and

$$\|\mathbf{v}_i\| = 1 \qquad \text{for every } i.$$

Equivalently,

$$\boxed{\langle \mathbf{v}_i, \mathbf{v}_j \rangle = \delta_{ij} = \begin{cases} 1, & i = j, \\ 0, & i \neq j, \end{cases}}$$

where δ_{ij} is the Kronecker delta.

Every orthogonal set of nonzero vectors is linearly independent

Key fact

Every orthogonal set of nonzero vectors is linearly independent.

Therefore, if $\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ is an orthogonal set and each $\mathbf{v}_i \neq \mathbf{0}$, then

$$c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k = \mathbf{0} \quad \Longrightarrow \quad c_1 = \cdots = c_k = 0.$$

Useful consequence

n nonzero orthogonal vectors in $\mathbb{R}^n$ automatically form a basis for $\mathbb{R}^n$.

Orthogonal coordinates are found one at a time

If $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is an orthogonal basis and

$$\mathbf{x} = c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n,$$

then

$$\boxed{c_i = \frac{\langle \mathbf{x}, \mathbf{v}_i \rangle}{\langle \mathbf{v}_i, \mathbf{v}_i \rangle}.}$$

For an orthonormal basis, this simplifies to

$$\boxed{c_i = \langle \mathbf{x}, \mathbf{v}_i \rangle.}$$

No simultaneous system is required: orthogonality isolates each coefficient.

Orthogonal matrices

A square matrix Q is orthogonal when its columns are orthonormal:

$$\boxed{Q^\mathsf{T} Q = I.}$$

Therefore

$$\boxed{Q^{-1} = Q^\mathsf{T}.}$$

For all vectors $\mathbf{x}, \mathbf{y}$,

$$\langle Q\mathbf{x}, Q\mathbf{y} \rangle = \mathbf{x}^\mathsf{T} Q^\mathsf{T} Q \mathbf{y} = \mathbf{x}^\mathsf{T} \mathbf{y}.$$

So orthogonal transformations preserve inner products, norms, angles and distances.
Rotations and reflections are the standard examples.

The spectral theorem gives an orthogonal diagonalisation

For every real symmetric matrix A, there are an orthogonal matrix Q and a real diagonal matrix D such that

$$\boxed{A = QDQ^{\mathsf{T}}.}$$

- The diagonal entries of D are the eigenvalues of A.
- The columns of Q are matching normalised eigenvectors.
- All eigenvalues are real.
- Eigenvectors can be chosen to form an orthonormal basis.

Compared with ordinary diagonalisation

$A = VDV^{-1}$ becomes $A = QDQ^{\mathsf{T}}$ because $Q^{-1} = Q^{\mathsf{T}}$.

Positive definite matrices

A real symmetric matrix A is positive definite when

$$\boxed{\mathbf{x}^\mathsf{T} A\mathbf{x} > 0 \quad \text{for every } \mathbf{x} \neq \mathbf{0}.}$$

The key eigenvalue test is

$$\boxed{A \text{ is positive definite} \iff \text{every eigenvalue of } A \text{ is positive.}}$$

Using $A = QDQ^\mathsf{T}$ and $\mathbf{y} = Q^\mathsf{T}\mathbf{x}$,

$$\mathbf{x}^\mathsf{T} A\mathbf{x} = \mathbf{y}^\mathsf{T} D\mathbf{y} = \lambda_1 y_1^2 + \cdots + \lambda_n y_n^2.$$

A non-diagonal matrix can be positive definite

Consider the symmetric matrix

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \qquad \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}.$$

Its quadratic form is

$$\begin{aligned} \mathbf{x}^\mathsf{T} A\mathbf{x} &= 2x_1^2 + 2x_1x_2 + 2x_2^2 \\ &= x_1^2 + x_2^2 + (x_1 + x_2)^2 > 0 \end{aligned}$$

for every $\mathbf{x} \neq \mathbf{0}$.

Eigenvalue check

The eigenvalues are 1 and 3. Both are positive, confirming that A is positive definite.

Least squares minimises the residual

When $X\mathbf{k} = \mathbf{y}$ is inconsistent, define the residual

$$\mathbf{r} = \mathbf{y} - X\mathbf{k}.$$

Least squares chooses $\mathbf{k}$ to minimise

$$\|\mathbf{r}\|^2 = \|\mathbf{y} - X\mathbf{k}\|^2.$$

At the best fit, the residual is orthogonal to every column of X:

$$\boxed{X^{\mathsf{T}}(\mathbf{y} - X\mathbf{k}) = \mathbf{0}.}$$

These are the normal equations.

The least-squares solution

Rearranging

$$X^{\mathsf{T}}(\mathbf{y} - X\mathbf{k}) = \mathbf{0}$$

gives

$$X^{\mathsf{T}}X\mathbf{k} = X^{\mathsf{T}}\mathbf{y}.$$

If $X^{\mathsf{T}}X$ is invertible, then

$$\boxed{\mathbf{k}_0 = (X^{\mathsf{T}}X)^{-1}X^{\mathsf{T}}\mathbf{y}.}$$

At the solution, the residual $\mathbf{y} - X\mathbf{k}_0$ is orthogonal to every column of X.

Least squares fits a line to inconsistent data

Fit $y = mx + c$ to $(0, 1)$, $(1, 2)$ and $(2, 2)$.

Set up the system

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 1 \\ 2 & 1 \end{pmatrix}, \quad \mathbf{k} = \begin{pmatrix} m \\ c \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}.$$

$$\boxed{X\mathbf{k} = \mathbf{y}}$$

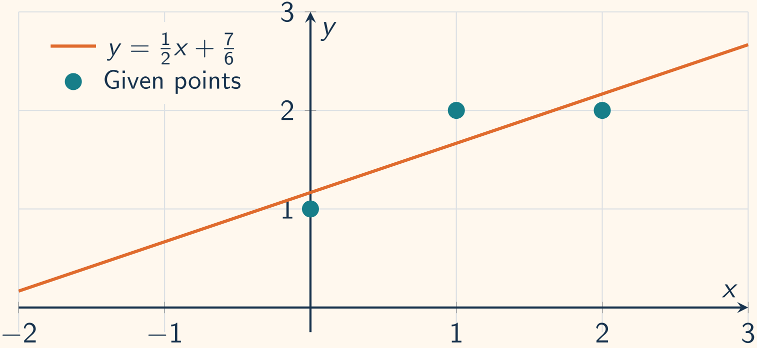
Least-squares solution

$$\mathbf{k}_0 = (X^\mathsf{T} X)^{-1} X^\mathsf{T} \mathbf{y}$$

$$\mathbf{k}_0 = \begin{pmatrix} 1/2 \\ 7/6 \end{pmatrix}$$

$$\boxed{y = \tfrac{1}{2}x + \tfrac{7}{6}}$$

The least-squares line approximates all three points



Short Revision Quiz

Inner products, orthogonality and least squares

Question 1 — Dot product

For $\mathbf{x} = (2, -1, 3)^\top$ and $\mathbf{y} = (1, 4, 2)^\top$, what is $\mathbf{x}^\top \mathbf{y}$?

A. 4

B. 8

C. 12

Question 2 — Norm

What is the Euclidean norm of $\mathbf{x} = (3, -4)^\top$?

A. 1

B. 5

C. 7

Question 3 — Inner product axioms

Which condition must every real inner product satisfy?

A. $\langle \mathbf{x}, \mathbf{x} \rangle < 0$ for nonzero $\mathbf{x}$

B. $\langle \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{y}, \mathbf{x} \rangle$

C. $\langle \mathbf{x}, \mathbf{y} \rangle$ is a vector

Question 4 — Weighted orthogonality

Under

$$\langle \mathbf{x}, \mathbf{y} \rangle = 2x_1y_1 + x_2y_2,$$

which vector is orthogonal to $(1, 2)^\mathsf{T}$?

A. $(2, -1)^\mathsf{T}$

B. $(1, -1)^\mathsf{T}$

C. $(1, 1)^\mathsf{T}$

Question 5 — Inner products of functions

On $C[0,1]$, let

$$\langle f, g\rangle = \int_0^1 f(x)g(x)\,\mathrm{d}x.$$

What is $\langle 1, x\rangle$?

A. $\dfrac{1}{2}$

B. 1

C. 0

Question 6 — Orthonormal sets

If $\{\mathbf{u}_1, \mathbf{u}_2\}$ is orthonormal, which statement is true?

A. $\mathbf{u}_1^\top \mathbf{u}_2 = 1$

B. $\|\mathbf{u}_1\| = \|\mathbf{u}_2\| = 0$

C. $\mathbf{u}_1^\top \mathbf{u}_2 = 0$ and both norms equal 1

Question 7 — Orthogonal matrices

If Q is orthogonal, which identity holds?

A. $Q^{-1} = Q^{\mathsf{T}}$

B. $Q^{-1} = Q$

C. $Q^{\mathsf{T}}Q = 0$

Question 8 — Spectral theorem

Which factorisation is guaranteed for a real symmetric matrix A?

A. $A = QDQ^\top$ with Q orthogonal

B. $A = Q + D$

C. $A = Q^\top Q$ for every symmetric A

Question 9 — Positive definiteness

A real symmetric matrix is positive definite exactly when:

A. All diagonal entries are positive

B. All eigenvalues are positive

C. Its determinant is negative

Question 10 — Least squares

At a least-squares solution, the residual $\mathbf{r} = \mathbf{y} - X\mathbf{k}$ is:

A. Orthogonal to every column of X

B. Equal to every column of X

C. Always the zero vector

Quiz answers 1–5

1 — A

$2(1) + (-1)(4) + 3(2) = 4.$

2 — B

$\sqrt{3^2 + (-4)^2} = 5.$

3 — B

Every real inner product is symmetric.

4 — B

$2(1)(1) + (2)(-1) = 0.$

5 — A

$$\langle 1, x \rangle = \int_0^1 x \, \mathrm{d}x = \left[\frac{x^2}{2} \right]_0^1 = \frac{1}{2}.$$

Quiz answers 6–10

6 — C

Orthonormal means mutually orthogonal unit vectors.

7 — A

$Q^{\mathsf{T}}Q = I$, so $Q^{-1} = Q^{\mathsf{T}}$.

8 — A

The spectral theorem gives $A = QDQ^{\mathsf{T}}$.

9 — B

A real symmetric matrix is positive definite iff all its eigenvalues are positive.

10 — A

The normal equations say $X^{\mathsf{T}}\mathbf{r} = \mathbf{0}$.

Final takeaway

Inner products define geometry; orthogonality simplifies coordinates; and least squares uses orthogonality to find the best approximation.